MSG No. 123.340 $P4/mmm1'$ [Type II, tetragonal]

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det	TR
1	$\{1\|0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
2	$\{4^+_{001}\|0\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
3	$\{4^-_{001}\|0\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
4	$\{2_{100}\|0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	1
5	$\{2_{010}\|0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	1
6	$\{2_{001}\|0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
7	$\{2_{110}\|0\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	1
8	$\{2_{1-10}\|0\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	1
9	$\{-1\|0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	1
10	$\{-4^+_{001}\|0\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	1
11	$\{-4^-_{001}\|0\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	1
12	$\{m_{100}\|0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	−1	1
13	$\{m_{010}\|0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	−1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
14	$\{\mathrm{m}_{001}\|0\}$	$\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&-1&0\end{bmatrix}$	-1	1
15	$\{\mathrm{m}_{110}\|0\}$	$\begin{bmatrix}0&-1&0&0\\-1&0&0&0\\0&0&1&0\end{bmatrix}$	-1	1
16	$\{\mathrm{m}_{1-10}\|0\}$	$\begin{bmatrix}0&1&0&0\\1&0&0&0\\0&0&1&0\end{bmatrix}$	-1	1
17	$\{1'\|0\}$	$\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\end{bmatrix}$	1	-1
18	$\{4^+_{001}{}'\|0\}$	$\begin{bmatrix}0&-1&0&0\\1&0&0&0\\0&0&1&0\end{bmatrix}$	1	-1
19	$\{4^-_{001}{}'\|0\}$	$\begin{bmatrix}0&1&0&0\\-1&0&0&0\\0&0&1&0\end{bmatrix}$	1	-1
20	$\{2_{100}{}'\|0\}$	$\begin{bmatrix}1&0&0&0\\0&-1&0&0\\0&0&-1&0\end{bmatrix}$	1	-1
21	$\{2_{010}{}'\|0\}$	$\begin{bmatrix}-1&0&0&0\\0&1&0&0\\0&0&-1&0\end{bmatrix}$	1	-1
22	$\{2_{001}{}'\|0\}$	$\begin{bmatrix}-1&0&0&0\\0&-1&0&0\\0&0&1&0\end{bmatrix}$	1	-1
23	$\{2_{110}{}'\|0\}$	$\begin{bmatrix}0&1&0&0\\1&0&0&0\\0&0&-1&0\end{bmatrix}$	1	-1
24	$\{2_{1-10}{}'\|0\}$	$\begin{bmatrix}0&-1&0&0\\-1&0&0&0\\0&0&-1&0\end{bmatrix}$	1	-1
25	$\{-1'\|0\}$	$\begin{bmatrix}-1&0&0&0\\0&-1&0&0\\0&0&-1&0\end{bmatrix}$	-1	-1
26	$\{-4^+_{001}{}'\|0\}$	$\begin{bmatrix}0&1&0&0\\-1&0&0&0\\0&0&-1&0\end{bmatrix}$	-1	-1
27	$\{-4^-_{001}{}'\|0\}$	$\begin{bmatrix}0&-1&0&0\\1&0&0&0\\0&0&-1&0\end{bmatrix}$	-1	-1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
28	$\{m_{100}'\|0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	-1	-1
29	$\{m_{010}'\|0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	-1	-1
30	$\{m_{001}'\|0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	-1	-1
31	$\{m_{110}'\|0\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	-1	-1
32	$\{m_{1-10}'\|0\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	-1	-1